

Expert Review Overview of the Manuscript

1 Overall Assessment

The manuscript demonstrates substantial technical merit and publication potential. Its primary weakness is not the underlying technical contribution, but rather how that contribution is framed and presented. The work presently reads as a technical validation report rather than a journal article.

- Technical Merit: High
- Novelty: High if framed as a physically validated optical simulation framework.
- Writing Quality: Requires substantial revision.
- Validation Methodology: Strong.
- Publication Potential: Strong after major revision.

2 Most Important Recommendation

The manuscript should not primarily be sold as “a QKD simulator.” Instead, it should be presented as an open-source, literature-traceable, physically validated optical channel simulation framework for QKD research. The component-wise validation methodology is the strongest contribution of this work.

3 Major Revisions Required

3.1 1. Reframe the Novelty

The primary contributions should be:

1. Literature-traceable physical-layer optical modelling.
2. Independent component-wise validation against published literature.
3. Unified complex-envelope formulation representing both polarization and optical power.
4. Fully reproducible validation pipeline and open-source implementation.
5. System-level demonstrations showing physically sensible degradation under combined impairments.

3.2 2. Expand the Literature Review

The literature review is currently too short. It should discuss:

- Protocol-level simulators (NetSquid, SimulaQron, QuTIP, SeQUeNCe, QuNetSim).
- Commercial optical simulators (OptiSystem, Lumerical, VPIphotonics and RSoft).
- Existing QKD simulators available in literature.
- Limitations of existing frameworks.

A comparison table is highly recommended. Suggested comparison metrics include:

- Physical-layer modelling
- Open-source availability
- Chromatic dispersion modelling
- PMD modelling
- Birefringence modelling
- Independent validation methodology

3.3 3. Improve the Introduction

The introduction should be approximately doubled in size and discuss:

- Quantum communication systems.
- Why physical-layer modelling matters for QKD.
- Environmental sensitivity studies.
- Limitations of existing simulators.
- Reproducibility concerns in simulation studies.
- Motivation for literature-traceable models.

3.4 4. Expand the System-Level Demonstration

The current system demonstration section is too small relative to the validation sections.

Recommended additions include:

- QBER versus distance.
- QBER versus pulse width.
- Temperature sensitivity analysis.
- PMD coefficient sensitivity analysis.
- Bend-induced birefringence studies.
- Combined impairment studies.

Potential scenarios include:

1. Distance variations (0–150 km).

2. PMD enabled versus disabled.
3. Temperature variations.
4. Fiber bend radius variations.
5. All impairments simultaneously enabled.

3.5 5. Presentation of Validation Results

The manuscript repeatedly uses phrases such as:

- Machine precision.
- Exact agreement.
- $R^2 = 1$.
- Errors smaller than $10^{-13}\%$.

Although mathematically reasonable for analytical validations, reviewers are frequently skeptical of overly perfect results. More appropriate wording would be:

Agreement was observed within floating-point precision owing to the analytical nature of the validation procedure.

This maintains accuracy while improving reviewer perception.

4 Strengths of the Work

The strongest aspects of the manuscript are:

- The single source of truth field formulation.
- Literature-traceable equations throughout the framework.
- Independent validation of every major component.
- Explicit reproducibility practices.
- Well-written limitations section.
- Open-source implementation.

5 Title Recommendations

Consider replacing the current title with one of the following:

1. An Open-Source Physically Validated Optical Channel Framework for Quantum Key Distribution.
2. A Literature-Traceable Physical-Layer Optical Framework for Quantum Key Distribution Simulation.
3. An Open-Source Physical-Layer Optical Simulation Framework for Quantum Key Distribution with Component-Wise Literature Validation.

The term “Framework” better reflects the contribution than “Model.”

6 Publication Strategy

The manuscript should be positioned as a validation and simulation framework paper rather than as a BB84 or QKD simulation paper. The validation methodology is the primary contribution, whereas the BB84 demonstration should be treated as evidence that the independently validated components compose correctly.

7 Final Verdict

The technical foundation is considerably stronger than the current presentation suggests. Assuming that the validation scripts and accompanying source code produce the reported results, this work possesses strong journal potential. The most significant improvements required involve presentation, literature coverage, framing of the contribution, and expansion of the system-level demonstrations. These are revision issues rather than fundamental technical deficiencies.

A major revision is recommended prior to submission. The strongest publication strategy is to emphasize physical-layer validation, literature traceability, reproducibility, and open-source availability as the central contributions of the work.